

SIMATS ENGINEERING
CO4 – AT2 – Capstone Project - Medium

COURSE CODE: ECA5613 / **COURSE NAME:** 5G / 6G Communication Systems
Name : **Slot: B**
Register No: P. Niharika **Duration: 120 Mins**
Assessment Tool Weightage: 35% **Total Marks:100**

QUESTIONS:1

Capstone Project Title : IoT-Based V2V Communication Alert System for Future 5G-Enabled Vehicle

Module Title: Future 5G Communication Integration Module

RUBRICS FOR CALCULATION:

Capstone Project					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25%	Explains concepts accurately with complete clarity; demonstrates deep understanding	Explains concepts correctly with minor gaps	Partial understanding; explanations contain errors or omissions	Unable to explain basic concepts
Application	25%	Applies concepts correctly to new situations; solves problems logically and independently	Applies concepts to familiar problems with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20%	Analyzes scenarios critically; provides strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis; weak justification	No analytical reasoning demonstrated
Communication	15%	Communicates ideas clearly, logically, and confidently with proper terminology	Communicates clearly with minor hesitation or lack of precision	Communication is unclear or lacks structure	Unable to communicate ideas effectively
Response to Questions	15%	Responds accurately, confidently, and extends answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions

Code of Conduct:

I __P. Niharika(192412226)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Niharika

Signature of the student:



PROBLEM STATEMENT



- Road accidents often occur due to **delayed driver response and limited awareness.**
- Traditional safety systems mainly monitor conditions **within a single vehicle.**
- V2V communication allows vehicles to **share speed, position, and warning information.**
- IoT enables **real-time collection and exchange of vehicle data.**
- 5G technology can provide **fast, reliable, and low-latency communication** for future smart transportation.



FUTURE SCOPE & RESULTS



- Future integration of 5G communication
- Faster and reliable vehicle-to-vehicle communication
- Support for intelligent transportation systems
- Cloud-based vehicle data storage and analysis
- Real-time traffic and emergency information sharing
- Improved system reliability, accuracy, and performance
- Quick emergency response during critical situations



Module 3: Future 5G Communication Integration Module



- **IoT Connectivity** – Connects the vehicle to the internet for real-time data exchange.
- **Wireless Communication** – Uses 4G/5G, Wi-Fi, or Bluetooth to transmit vehicle information.
- **Vehicle-to-Vehicle (V2V)** – Enables vehicles to share safety and traffic information.
- **Cloud Communication** – Sends vehicle data to cloud platforms for storage and analysis.
- **Real-Time Alerts** – Provides instant notifications about accidents, faults, traffic, or emergencies.

